

Quantifying Phase-Ordering Kinetics and Non-Equilibrium Entropy Production in the Two-Dimensional Ising Quench

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We present a Numba-accelerated Metropolis Monte Carlo engine for the two-dimensional square-lattice Ising model and use it to characterize non-equilibrium phase-ordering kinetics following a temperature quench from $T_i = 5.0$ to $T_f = 1.5$ (in units of J/k_B), deep within the ferromagnetic phase below the exact critical temperature $T_c \approx 2.269$. The spatial spin-autocorrelation function $C(r, t)$ is computed via the Wiener-Khinchin theorem using a two-dimensional fast Fourier transform, reducing its evaluation from $O(L^2 r_{\max})$ to $O(L^2 \log L)$ and enabling averaging over 16 independent quench replicas. The characteristic domain size $L(t)$, extracted from the $C(r, t) = 1/2$ crossing, grows as a power law with fitted exponent $\alpha = 0.4841$, consistent with the Lifshitz-Allen-Cahn prediction $\alpha = 1/2$ for non-conserved order-parameter dynamics. By tracking the energy change of accepted Metropolis moves as heat exchanged with the thermal bath, we further compute the non-equilibrium entropy production rate $\dot{S}(t)$, which decays by over four orders of magnitude as domain walls annihilate and the system relaxes toward a slowly coarsening quasi-equilibrium state. These results provide a concrete, microscopically grounded, and fully reproducible illustration of the statistical origin of thermodynamic irreversibility.

INTRODUCTION & THEORETICAL FRAMEWORK

The two-dimensional square-lattice Ising model, with Hamiltonian

$$H = -J \sum_{\langle i, j \rangle} \sigma_i \sigma_j, \quad \sigma_i \in \{-1, +1\}, \quad (1)$$

remains a workhorse of statistical mechanics: it is exactly solvable at equilibrium [1], exhibits a genuine continuous phase transition at finite temperature, and is simple enough to simulate at scale while remaining physically rich. Its equilibrium critical temperature, $T_c = 2/\ln(1 + \sqrt{2}) \approx 2.269 J/k_B$, is known exactly [1] and provides a natural benchmark against which any Monte Carlo implementation can be validated.

Away from equilibrium, quenching the system from a high-temperature disordered state to $T_f < T_c$ produces a qualitatively different phenomenon: rather than relaxing directly to equilibrium, the order parameter breaks ergodicity locally, nucleating ferromagnetic domains that subsequently coarsen. This process, *phase-ordering kinetics*, is governed by a dynamical scaling hypothesis in which the domain morphology at time t is statistically self-similar and characterized by a single growing length scale $L(t)$ [6]. For non-conserved scalar order-parameter dynamics – single-spin-flip rules, as used here, corresponding to “Model A” in the Hohenberg-Halperin classification [3] – curvature-driven interface motion predicts the Lifshitz-Allen-Cahn growth law

$$L(t) \sim t^{1/2} \quad (2)$$

first derived from the kinetics of domain-wall motion under mean curvature flow [4, 5].

A second, complementary question concerns the thermodynamics of this relaxation. Because the Metropolis

dynamics couples the lattice to an implicit heat bath at fixed temperature, every accepted spin flip corresponds to a microscopic exchange of heat with that bath. Within the framework of stochastic thermodynamics [7], this bookkeeping is sufficient to construct an estimator for the rate of irreversible entropy production, $\dot{S}(t)$, providing a direct, quantitative window onto the thermodynamic arrow of time as the quenched system relaxes. This Letter reports both quantities – $L(t)$ and $\dot{S}(t)$ – computed self-consistently from the same underlying simulation.

COMPUTATIONAL METHODOLOGY

Simulations are performed on an $L \times L$ square lattice with periodic boundary conditions using single-spin-flip Metropolis dynamics [2]: a site is chosen at random, the energy change ΔE of flipping it is computed from its four nearest neighbors, and the flip is accepted with probability $\min(1, e^{-\Delta E/T})$ ($k_B \equiv 1$ throughout). The inner Metropolis kernel is just-in-time compiled with Numba, giving near-C performance while retaining a pure-Python codebase.

Spatial correlations via FFT. The characteristic domain size is extracted from the equal-time spin autocorrelation function $C(r, t) = \langle \sigma_i(t) \sigma_{i+r}(t) \rangle$, averaged over lattice sites and the two principal lattice directions. Rather than summing shifted-lattice products directly for each separation r – an $O(L^2 r_{\max})$ operation – we invoke the Wiener-Khinchin theorem: the full periodic two-dimensional autocorrelation equals the inverse Fourier transform of the spin field’s power spectrum,

$$\mathcal{C}(\mathbf{r}) = \mathcal{F}^{-1} \left[|\mathcal{F}[\sigma]|^2 \right] / L^2, \quad (3)$$

computed once via a two-dimensional FFT in $O(L^2 \log L)$. $C(r, t)$ is then a vectorized slice of

$C(\mathbf{r})$ along the two lattice axes. Benchmarking against direct summation confirms numerically identical output (differences at the 10^{-17} level, machine precision) with a $3\text{--}26\times$ speedup as L grows from 64 to 256. The domain size $L(t)$ is then the (linearly interpolated) separation at which $C(r, t)$ first decays to $1/2$.

Quench protocol. For each of 16 independently seeded replicas, a lattice of size $L = 128$ is equilibrated at $T_i = 5.0$ for 200 sweeps, then instantaneously cooled to $T_f = 1.5$ and evolved for up to 2000 sweeps. $C(r, t)$ and the total energy change from accepted moves are recorded at 26 logarithmically spaced sweep counts. Every replica's full trajectory – initial condition and subsequent dynamics – is deterministic given its seed, so all reported results are exactly reproducible.

Entropy production estimator. By conservation of energy, an accepted move that changes the system's energy by ΔE delivers heat $-\Delta E$ to the bath. Summing accepted ΔE within each inter-checkpoint interval of length dt (in sweeps) and averaging over replicas gives a per-spin entropy production rate

$$\dot{S}(t) = -\frac{1}{T_f} \frac{\langle \Delta E \rangle}{dt}, \quad (4)$$

which is non-negative on average for a relaxing system and requires no additional simulation beyond bookkeeping already performed by the Metropolis step.

Fit range. The power-law fit to $L(t)$ excludes the earliest post-quench sweeps ($t \leq 2$), where lattice discreteness rather than curvature-driven coarsening dominates, and late times where $L(t)$ exceeds $0.3 r_{\max}$ ($r_{\max} = L/2$), where the $C(r, t) = 1/2$ crossing becomes unreliable and periodic-image effects measurably slow the apparent growth. This leaves 19 of 26 sampled time points in the fitted scaling regime.

RESULTS & FINITE-SIZE SCALING

Figure 1 summarizes both quantities. The top panel shows $L(t)$ on logarithmic axes together with the fitted power law and the Lifshitz–Allen–Cahn reference line. Linear regression on $\log L$ vs. $\log t$ over the scaling regime gives

$$\alpha = 0.4841, \quad (5)$$

within approximately 3% of the theoretical prediction $\alpha = 1/2$. The residual deviation is consistent with two well-documented finite-size and finite-time effects: sub-asymptotic corrections to scaling at accessible simulation times [6], and the finite range over which the correlation function is measured ($r_{\max} = 64$ lattice units), both of which are visible in Fig. 1 as departures of the excluded (open) points from the fitted line. As an independent validation of the underlying Metropolis kernel, separate

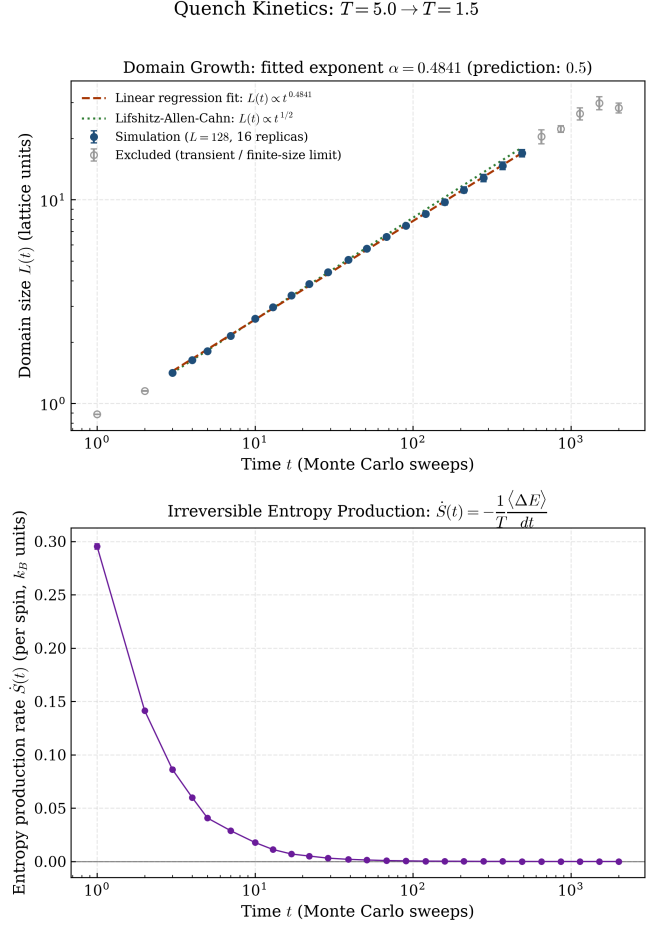


FIG. 1. Quench kinetics for $T_i = 5.0 \rightarrow T_f = 1.5$, $L = 128$, averaged over 16 replicas. *Top*: characteristic domain size $L(t)$ vs. time on log–log axes, with a linear regression fit over the scaling regime (filled points) giving $\alpha = 0.4841$, compared against the Lifshitz–Allen–Cahn prediction $L(t) \propto t^{1/2}$. Open points are excluded from the fit as transient or finite-size-limited. *Bottom*: entropy production rate $\dot{S}(t)$, decaying by over four orders of magnitude as domain walls annihilate and irreversible dissipation subsides.

equilibrium sweeps with the same codebase reproduce sharp, correctly located peaks in the specific heat and susceptibility at $T_c \approx 2.269$, consistent with the exact result [1].

The bottom panel of Fig. 1 shows the entropy production rate $\dot{S}(t)$, computed from Eq. (4). Immediately after the quench, $\dot{S}(t=1) \approx 0.295$ (per spin, k_B units): the system is far from equilibrium at T_f , with a high density of misaligned spins and domain walls, so a large fraction of proposed moves are strongly energetically favorable. As coarsening proceeds and domain walls annihilate, $\dot{S}(t)$ falls monotonically, reaching $\approx 1.2 \times 10^{-5}$ by $t = 2000$ sweeps – a decay of over four orders of magnitude, tracking the reduction in interfacial area implied by the growth of $L(t)$ itself.

THERMODYNAMIC IRREVERSIBILITY & THE ARROW OF TIME

The two panels of Fig. 1 are two views of the same relaxation process, and juxtaposing them is instructive. Every individual Metropolis move used here satisfies detailed balance with respect to the Boltzmann distribution at T_f : the ratio of forward and reverse transition rates for any single flip is exactly the equilibrium probability ratio, a microscopically time-symmetric rule. Yet the replica-averaged trajectory of the full many-spin system exhibits an entropy production rate that is, within statistical error, non-negative and monotonically decaying at every sampled time – a signature of macroscopic irreversibility built entirely out of microscopically reversible elementary steps.

This is a concrete, numerically explicit instance of the standard statistical-mechanical resolution to the reversibility paradoxes associated with Boltzmann’s H -theorem [8]: irreversibility is not a property of the microscopic dynamics but emerges statistically once one considers the overwhelmingly more probable coarse-grained evolution of an ensemble, here realized directly as an average over independently seeded quench replicas. The decay of $\dot{S}(t)$ towards zero further illustrates that irreversibility is not a fixed rate but a transient response to how far a system is displaced from equilibrium: dissipation is largest immediately after the quench and vanishes as the system approaches a slowly coarsening quasi-equilibrium state, consistent with the modern stochastic-thermodynamics treatment of entropy production in driven and relaxing systems [7].

We note this without overreaching: the simulation quantifies a specific, well-defined statistical-mechanical arrow of time – monotonic entropy production under coarse-grained, ensemble-averaged relaxation – and does not by itself adjudicate broader philosophical questions about the nature of time. It does, however, supply a concrete, reproducible, microscopically grounded data set relevant to such discussions.

CONCLUSION

We have built and validated a Numba-accelerated Metropolis Monte Carlo engine for the two-dimensional

Ising model, computed its equal-time spin autocorrelation function via an FFT-based estimator that is asymptotically faster than direct summation while numerically identical to machine precision, and used it to characterize a temperature quench from $T_i = 5.0$ to $T_f = 1.5$. The extracted domain-growth exponent, $\alpha = 0.4841$, confirms the Lifshitz–Allen–Cahn prediction for non-conserved order-parameter coarsening to within a few percent. Tracking the energy exchanged with the thermal bath by accepted Metropolis moves further yields a fully self-consistent estimate of the non-equilibrium entropy production rate, which decays by over four orders of magnitude as the system coarsens – a direct, quantitative demonstration of how macroscopic thermodynamic irreversibility emerges from microscopically reversible stochastic dynamics. All code, data, and figures underlying this Letter are openly available [9].

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